



COMP9444: Neural Networks and Deep Learning

Week 2b. Generalization

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September 16, 2024

Outline

- Supervised Learning
- Ockham's Razor
- Training, Validation and Test Error
- Dropout

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Types of Learning

- Supervised Learning
 - agent is presented with examples of inputs and their target outputs
- Reinforcement Learning
 - agent is not presented with target outputs, but is given a reward signal, which it aims to maximize
- Unsupervised Learning
 - agent is only presented with the inputs themselves, and aims to find structure in these inputs

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Supervised Learning

- we have a *training set* and a *test set*, each consisting of a set of items; for each item, a number of input attributes and a target value are specified.
- the aim is to predict the target value, based on the input attributes.
- agent is presented with the input and target output for each item in the training set; it must then predict the output for each item in the test set
- various learning paradigms are available:
 - Neural Network
 - Decision Tree
 - Support Vector Machine, etc.

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Supervised Learning – Issues

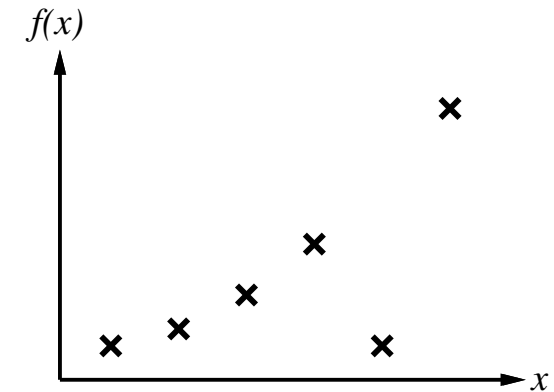
- framework (decision tree, neural network, SVM, etc.)
- representation (of inputs and outputs)
- pre-processing / post-processing
- training method (perceptron learning, backpropagation, etc.)
- generalization (avoid over-fitting)
- evaluation (separate training and testing sets)

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Curve Fitting

Which curve gives the “best fit” to these data?

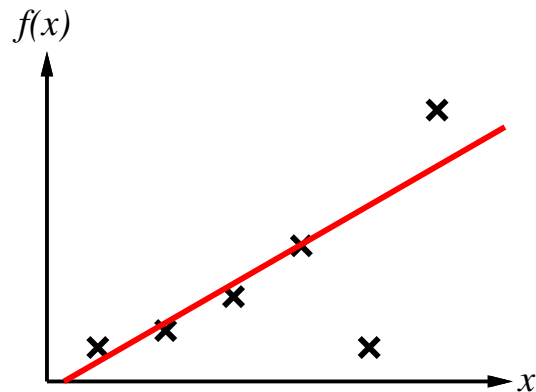


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Curve Fitting

Which curve gives the “best fit” to these data?



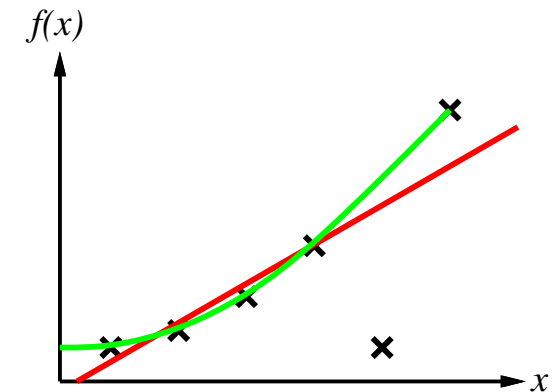
straight line?

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Curve Fitting

Which curve gives the “best fit” to these data?



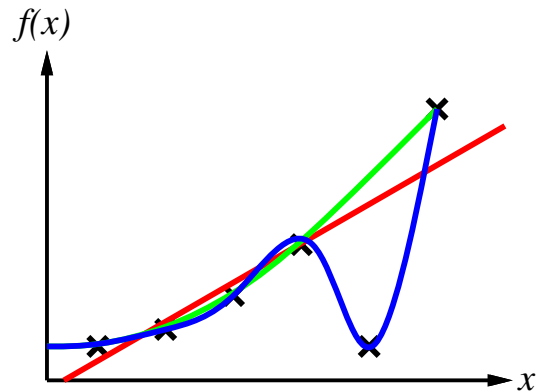
parabola?

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Curve Fitting

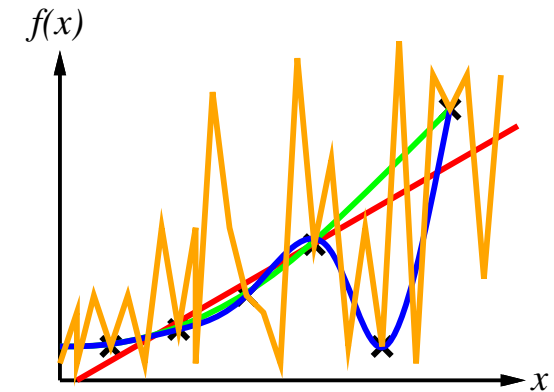
Which curve gives the “best fit” to these data?



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Curve Fitting

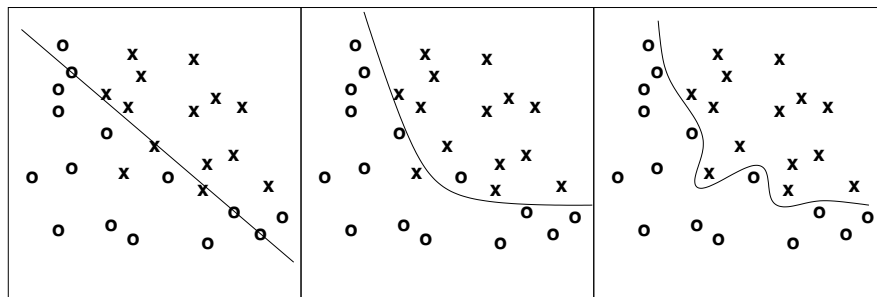
Which curve gives the “best fit” to these data?



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Ockham's Razor

“The most likely hypothesis is the *simplest* one consistent with the data.”



inadequate

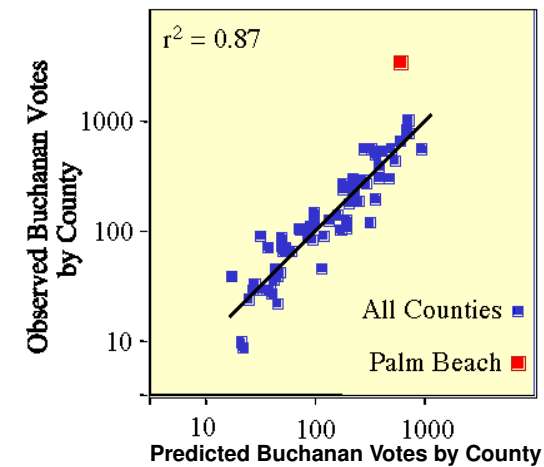
good compromise

over-fitting

Since there can be **noise** in the measurements, in practice we need to make a tradeoff between simplicity of the hypothesis and how well it fits the data.

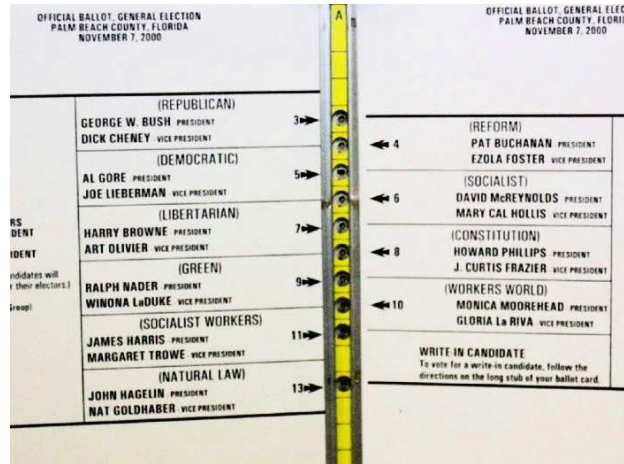
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Outliers



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Butterfly Ballot



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Ways to Avoid Overfitting in Neural Networks

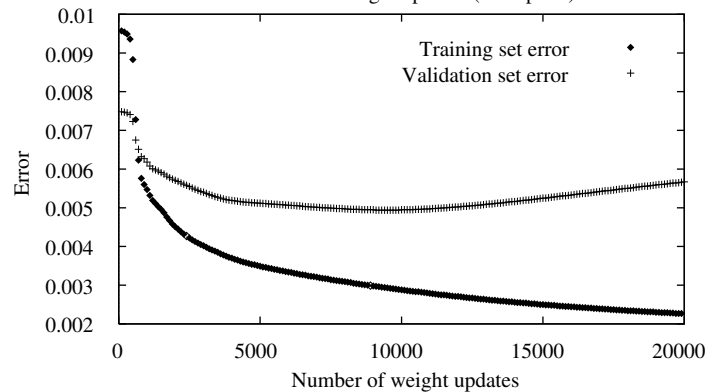
- Limit the number of hidden nodes or connections
- Limit the number of training epochs (weight updates)
- Dropout
- Weight Decay (Week 3)
- Data Augmentation (Week 4)
- Regularization

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Training, Validation and Test Error

Error versus weight updates (example 1)



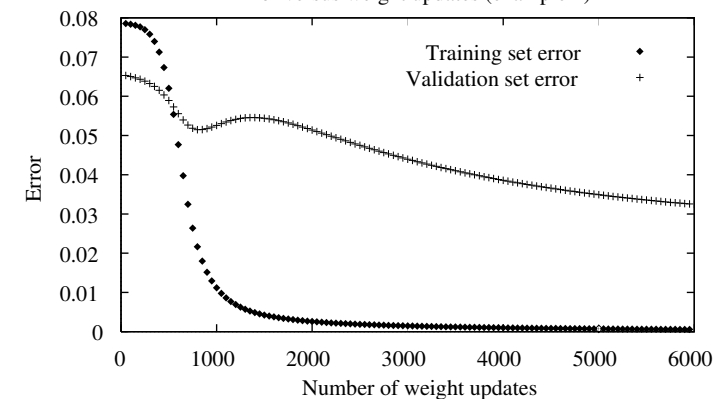
The x -axis could be the number of weight updates, hidden nodes, dropout, etc.
Normally, Training Error decreases, but Validation and Test Error flatter or increase.

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Be Careful about Early Stopping

Error versus weight updates (example 2)

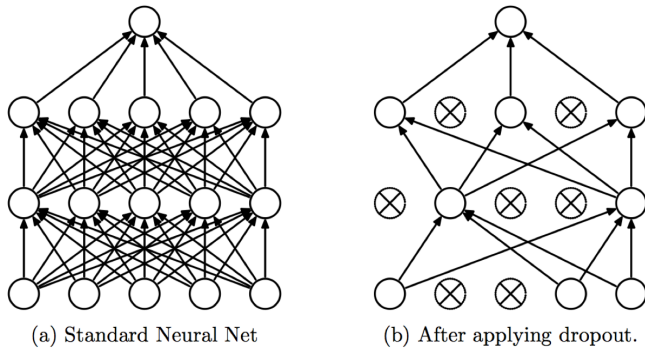


We may need to continue training in order to be sure what is going on.

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Dropout



For each minibatch, randomly choose a subset of nodes to not be used. Each node is chosen with some fixed probability (usually, one half).

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Dropout

- When training is finished and the network is deployed, all nodes are used, but the activation of each node is multiplied by the probability of “keeping” the node during training.
- Thus, the activation received by each node is the *average* value of the activation it would have received during training.
- Dropout forces the network to achieve *redundancy* because it must deal with situations where some features are missing.
- Another way to view dropout is that it implicitly (and efficiently) simulates an *ensemble* of different architectures.

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Ensembling

- Ensembling is a method where a number of different classifiers are trained on the same task, and the final class is decided by “voting” among them.
- In order to benefit from ensembling, we need to have *diversity* in the different classifiers.
- For example, we could train three neural networks with different architectures, three Support Vector Machines with different dimensions and kernels, as well as two other classifiers, and ensemble all of them to produce a final result. (Kaggle Competition entries are often done in this way).

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Dropout as an Implicit Ensemble

- In the case of dropout, the same data are used each time but a different architecture is created by removing the nodes that are dropped.
- The trick of multiplying the output of each node by the probability of dropout implicitly averages the output over all of these different models.

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